Haoran Wan

35 Olden Street, Princeton, NJ, 08540 ◇ Email: haoran.w@princeton.edu ◇ Mobile: 609 366 6317 ◇ Homepage ◇ Google Scholar

Education

- **Princeton University**, Ph.D. Candidate in Computer Science, Advisor: Kyle Jamieson — July 2023 - June 2028 (Anticipated)
  Teaching Assistant: Distributed Systems Fall ’25, Systems and Machine Learning Spring ’26
- **Nanjing University**, M.S. in Computer Science, Advisor: Wei Wang — Sept. 2019 - June 2023
  Outstanding Graduate Students Award ’21; Principal Special Scholarship ’19
- **University of Electronic Science and Technology of China**, B.Eng. in Network Engineering — Sept. 2015 - July 2019
  China National Scholarship ’17; 2nd People’s Scholarship ’16, ’18

Skills Summary

- **ML Systems:** LLM Serving (SGLang, HiCache, KV cache management), TensorRT.
- **Languages & GPU Programming:** C/C++, Python/PyTorch, CUDA, Triton, TMA, Nsight Systems/Compute.
- **Systems & Networking:** Linux Networking (AF_XDP, eBPF), Memory/Cache Subsystems, Congestion Control.

Industry Experience

NVIDIA | *Software Engineering Intern, Aerial (AI-RAN) Performance Team* — May 2026 - Aug. 2026

- **Aerial**: CUDA Software Performance Optimization | CUDA, CUTLASS/CuTe, TMA, Nsight, Memory/Cache Subsystem
  - Accelerated a production data-encoding CUDA kernel pipeline by **2.2x on Hopper and Blackwell GPUs** through kernel fusion, warp specialization, and fine-grained parallelism; merged into the Aerial codebase.
  - Sped up the memory copy by **2.5x** on Hopper/Blackwell GPUs by replacing the cp.async() with a pipelined **TMA** design built on CUTLASS/CuTe primitives.
  - Reduced tail-latency inflation by **63%** for multi-tenant workloads on GH200 by partitioning host CPU cores, L3 cache, and DRAM bandwidth with ARM MPAM.

Selected Research Projects

Machine Learning Systems

- **AgentCache (ongoing)**: Just-In-Time KV Cache Prefetch for Long-Context Agentic LLM Serving | SGLang, HiCache, KV Cache
  - Profiled SGLang HiCache serving Qwen3-30B on agentic workload: storage read queuing takes over **40%** of TTFT, as returning turns wait on NVMe/SSD fetches of evicted KV cache and blind polled retries cause read/write contention.
  - Designed a just-in-time prefetch scheme that uses framework-agnostic hints from the agent system to fetch each request’s prefix cache **before the agent’s next turn arrives** or as it nears the queue head, hiding storage latency and smoothing read bursts.
  - Replaced HiCache’s polled retries with an exactly-once storage fetch per request, easing storage read/write contention.
- **ALT (EuroSys ’23):** End-to-End Deep Learning Compiler | C++, CUDA, Apache TVM, ML Compiler
  - Engineered an auto-tuning engine on Apache TVM with a cost model that jointly searches best tensor layouts (NCHW/NHWC) and loop tiling strategies for ML model inference.
  - Co-designed an ML compiler that unifies graph-level data layout and operator-level loop optimizations, resolving the phase-ordering problem in traditional compilation stacks.
  - Achieved **1.5x** inference speedup on single operators (Conv2D, MatMul) and **1.4x** end-to-end speedup on ResNet, BERT, MobileNet models by generating hardware-aware CUDA/LLVM code.

Networking Systems

- **CellTwin (NSDI ’27) & NR-Scope (CoNEXT ’24)**: ML-Based Network Emulation | PyTorch, Linux, AF_XDP, C++
  - Designed and trained a transformer-based cellular network emulator for evaluating cross-domain network and application algorithms, cutting the emulation error by over **55%** compared with existing methods.
  - Served the real-time model with TensorRT and quantization inside a Linux AF_XDP middlebox, inferencing at **0.41 ms** latency.
- **L4Span (CoNEXT ’25)**: Ultra-Low Latency Network Stack Optimization | C++, Linux, Congestion Control
  - Engineered a **zero-copy** packet processing module in an open-source RAN, integrating L4S and ECN congestion signaling.
  - Cut queuing delay by **95%** at line-rate throughput for TCP Prague, BBRv2/v3, and SCReAM using a real-time capacity predictor that tunes queue management from sub-ms telemetry.

Selected Publications

- **[USENIX NSDI ’27]** CellTwin: Reproducible Cellular Network Emulation via High Resolution, in situ Measurement-Driven ML. **Haoran Wan**, Yaxiong Xie, Kyle Jamieson
- **[ACM SIGCOMM ’26]** Synchronizing with the Scheduler: Dual-Loop Congestion Control for 5G Uplink on Commodity Devices. Qiang Wu, Yuxin Liu, Tianyang Zhang, **Haoran Wan**, Kyle Jamieson, Yaxiong Xie
- **[ACM CoNEXT ’25]** L4Span: Spanning Congestion Signaling over NextG Networks for Interactive Applications. **Haoran Wan**, Kyle Jamieson
- **[ACM IMC ’25]** Automated, Cross-Layer Root Cause Analysis of 5G Video-Conferencing Quality Degradation. Fan Yi, **Haoran Wan**, Kyle Jamieson, Oliver Michel
- **[ACM CoNEXT ’24]** NR-Scope: A Practical 5G Standalone Telemetry Tool. **Haoran Wan**, Xuyang Cao, Alexander Marder, Kyle Jamieson
- **[ACM EuroSys ’23]** ALT: Boosting Deep Learning Performance by Breaking the Wall between Graph and Operator Level Optimizations. Zhiying Xu, Jiafan Xu, Hongding Peng, Wei Wang, Xiaoliang Wang, **Haoran Wan**, Haipeng Dai, Yixu Xu, Hao Cheng, Kun Wang, Guihai Chen